ELECTRICAL RECEPTACLE REPLACEMENT GUIDE

Part 1: Safety Protocols, Tools & Terminal Anatomy (120V / 15A-20A)

⚠CRITICAL SAFETY NOTICE (NFPA 70 / NEC COMPLIANCE)

Always switch off the circuit breaker at the main service panel before removing the wall plate. Never rely on a wall switch to de-energize an outlet. Verify with a tested non-contact voltage tester or multimeter.

REQUIRED TOOLS & SUPPLIES

- Non-contact voltage tester (NCVT)
- Digital multimeter or 3-light receptacle tester
- Wire strippers (12 & 14 AWG) with cutter
- Lineman's / needle-nose pliers
- #2 Phillips & flathead screwdrivers
- Replacement outlet (Tamper-Resistant / TR)
- Wire nuts / WAGO 221 lever nuts
- 6-inch solid copper pigtail jumpers (14/12 AWG)

5-STEP DE-ENERGIZATION PROTOCOL

1. Switch Breaker OFF
Locate panel and trip the matching branch breaker.

2. Test the Tester
Test NCVT on a known live circuit to confirm it lights up.

3. Probe Outlet Slots
Probe both slots of target outlet (must be silent).

4. Unscrew & Expose
Carefully pull outlet forward without touching terminals.

5. Probe Conductors
Verify zero voltage directly on all black & white screws.

STANDARD CONDUCTOR COLOR CODES

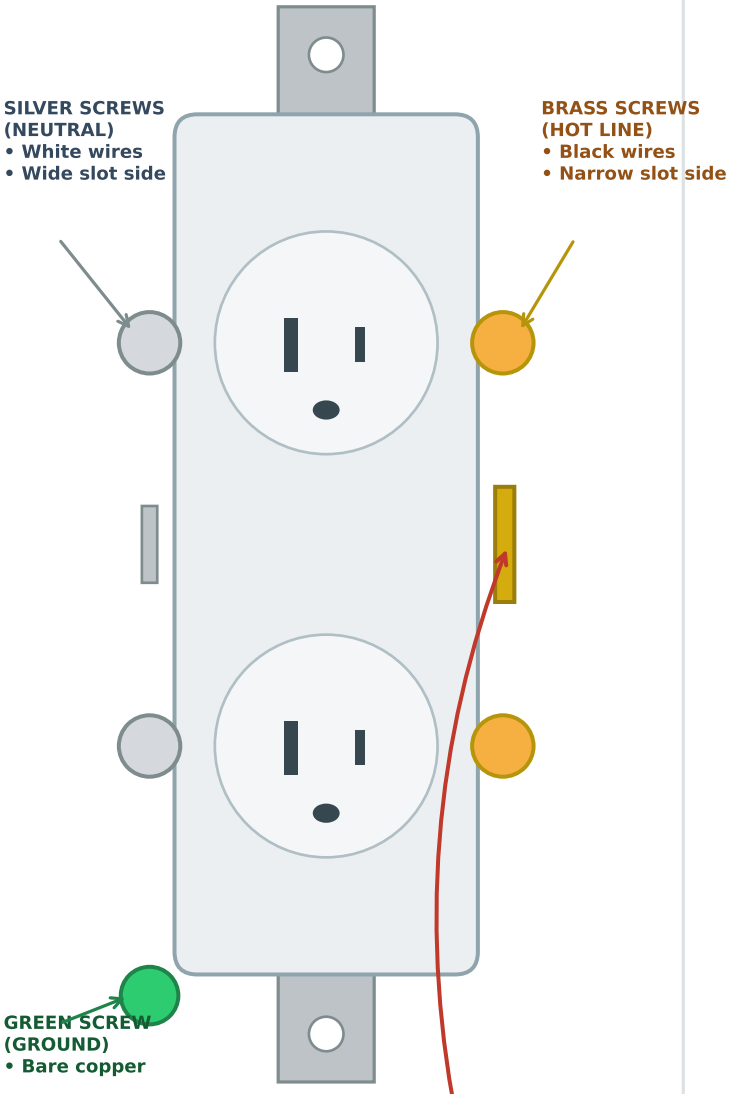
BLACK (Hot / Ungrounded)
120V potential from panel or switch.

WHITE (Neutral / Grounded)
Returns current to source (silver terminal).

BARE / GREEN (Grounding)
Safety path. Directs fault currents to earth.

RECEPTACLE TERMINAL ANATOMY

Understanding terminal polarities and mechanical break tabs



★ CRITICAL BREAK-OFF TAB

- INTACT: Both outlets share common hot line.
- REMOVED: Top & bottom are electrically split (used for wall-switch controlled half-hot outlets).

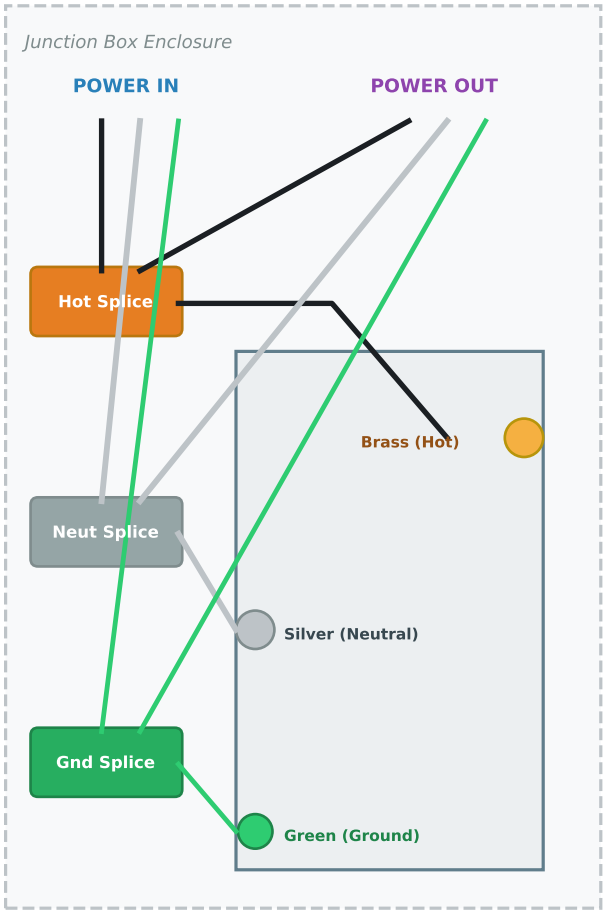
DAISY-CHAIN / FEED-THROUGH CONFIGURATION

Part 2: Two Cables Entering Box (2 Black, 2 White, 2 Bare Ground Wires)

WHY ARE THERE 2 BLACKS AND 2 WHITES?

Cable 1 brings 120V power IN from the panel or previous outlet. Cable 2 takes power OUT downstream to the next outlet. You must maintain circuit continuity across this box using one of two approved methods below.

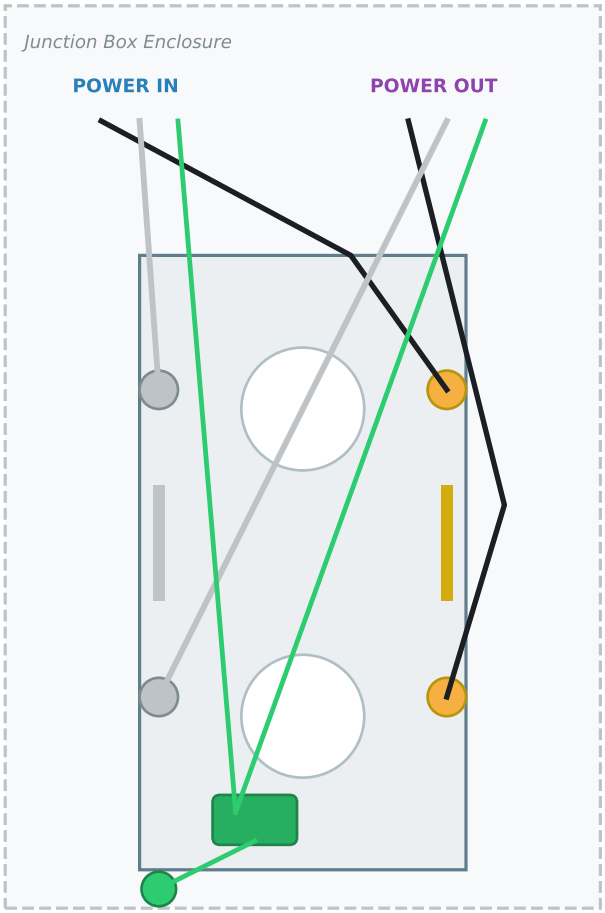
OPTION A: PIGTAILING (ELECTRICIAN STANDARD)



Key Benefits:

- ✔ Single conductor under each receptacle screw terminal.
- ✔ If this outlet burns out or is removed, downstream devices maintain power & neutral continuity.
- ✔ Complies with NEC 300.13(B) for multi-wire circuits.
- ✔ Folds cleanly into back of standard junction boxes.

OPTION B: DIRECT TERMINAL CONNECTION



Considerations & Trade-Offs:

- ✔ Uses less wire-fill volume in shallow/tight boxes.
- ✔ Faster to install when box depth is restricted.
- ✖ WARNING: Relies on receptacle metal tab to bridge power downstream. If outlet fails, downstream loses power.
- ✖ Never attach two wires under a single screw head.
- ✖ Do NOT use push-in 'back-stab' spring holes.

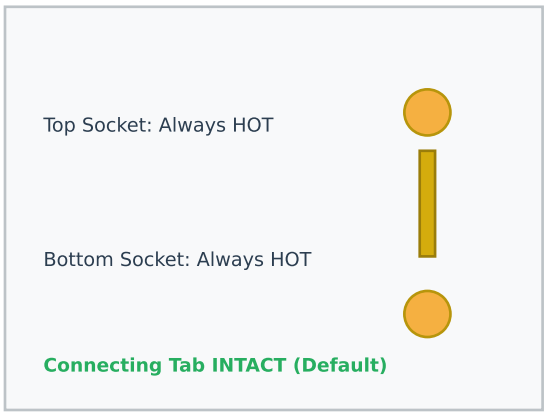
SPLIT-CIRCUITS & COMMISSIONING VERIFICATION

Part 3: Half-Hot Switched Outlets, Terminal Torquing & 3-Light Testing

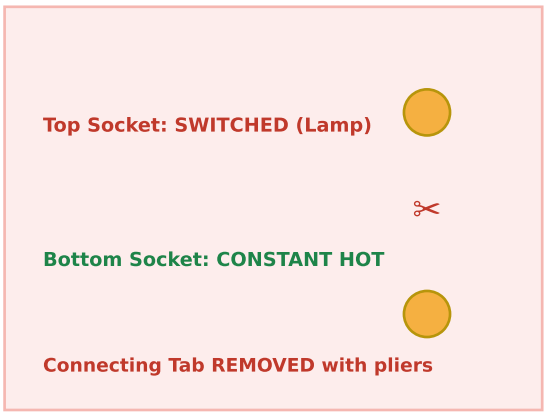
SPECIAL CASE: HALF-HOT (WALL-SWITCH CONTROLLED OUTLET)

In many living rooms and bedrooms, one receptacle socket is powered continuously for electronics, while the other is controlled by a wall switch for floor lamps. If your original outlet was configured this way:

Standard Duplex (Common Feed)



Split Duplex (Tab Removed / Half-Hot)



How to break tab: Grip the brass fin firmly with needle-nose pliers and wiggle it back and forth 2-3 times until it snaps clean off.

PROPER TERMINATION TECHNIQUE

- **Strip Length (3/4"):**
  Strip 3/4 inch of insulation without nicking the copper.
- **Clockwise J-Hook:**
  Bend bare copper into a tight J-hook. Hook around screw clockwise so tightening pulls the wire tighter into terminal.
- **No Insulation In Terminal:**
  Ensure no wire insulation gets trapped under the screw head, which causes arcing and overheating.
- **Torque to Spec:**
  Tighten terminal screws firmly (14-16 in-lbs torque). Loose screws are the #1 cause of residential electrical fires.
- **Fold Accordion-Style:**
  Push wires back accordion-style; do not let bare ground wire touch the hot brass terminal screws.



Tighten Clockwise ↻

Wire wraps 3/4 around shank in direction of screw turn.

TESTING & COMMISSIONING KEY

Plug in a 3-light receptacle tester to verify wiring:



CORRECT WIRING
Ready for use



OPEN GROUND
Check bare copper / green screw



OPEN NEUTRAL
Check white wires / silver screw



HOT / GROUND REV.
DANGER: Hot wired to ground!



HOT / NEUT REV.
Reverse polarity (Black on silver)

FINAL COMMISSIONING CHECKLIST:

[ ] Screws torqued (14-16 in-lbs) [ ] Grounds clear of screws
[ ] Cover plate secured [ ] Verified with 3-light tester